rocket engine

Y17 (2017) — English translation

In this problem we will study the flow of gas in the nozzle of a rocket engine, and then the flight of the rocket in airspace and in vacuum.

To begin with, let us recall several fundamental relationships that describe the behavior of flowing liquids or gases.

Part A. Dynamic relations for gas (1.0 points)

Let us consider a small cylinder of matter with density ρ (see figure), moving under the influence of a pressure difference. The movement is stationary and one-dimensional.

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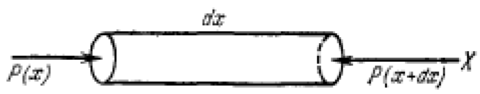

Fig. 1.

A1	Write down the equation of motion of the cylinder. Determine the pressure gradient dp/dx. Express your answer in terms of ρ, v and dv/dx.	**0.50**

A2	Determine the speed of sound c in the gas. Express your answer in terms of p, ρ and the adiabatic exponent γ.	**0.50**

Part B. Mach numbers and nozzle profile (2.0 points)

To describe gas flows, it is convenient to introduce the parameter Mach number $M = v/c$, which characterizes the ratio of the local gas flow velocity to the speed of sound.

We now move on to a qualitative description of the rocket engine nozzle. In a rocket engine, fuel burns in the combustion chamber and enters the nozzle. The speed of combustion products increases throughout the nozzle.

Solve the following points of the problem under the following assumptions: the gas is considered ideal; the gas flow is isentropic (that is, it has constant entropy, friction forces and dissipative losses are not taken into account) and adiabatic (that is, heat is not supplied or removed); the gas flow is stationary and one-dimensional, that is, at any fixed point of the nozzle, all flow parameters are constant in time and change only along the nozzle axis, and at all points of the selected cross section the flow parameters are the same, and the gas velocity vector is everywhere parallel to the symmetry axis of the nozzle; the gas mass flow rate is the same in all cross sections of the flow; the influence of all external forces and fields (including gravitational) is negligible; the symmetry axis of the nozzle is the spatial coordinate x.

Solve the following points of the problem under the following assumptions:

B1	Let us denote the cross-sectional area of the nozzle as A. Determine the value of dA/dx. Express your answer in terms of A, v, c and dv/dx.	**1.00**

B2	Qualitatively construct the profile of the nozzle section $A(x)$ when the gas velocity at the outlet is $v<c$. We can assume that at the entrance to the nozzle the gas velocity is negligible. Indicate special points (if any).	**0.50**

B3	Qualitatively construct the profile of the nozzle section $A(x)$ when the gas velocity at the outlet is $v>c$. We can assume that at the entrance to the nozzle the gas velocity is negligible. Indicate special points (if any).	**0.50**

Part C: Analysis of Nozzle Flow (5.0 points)

Let us consider in more detail the process of gas flow in the nozzle. Let the gas pressure p_c, temperature T_c and density ρ_c at the nozzle inlet. Gas adiabatic index γ.

C1	When making a rocket engine, the desired pressure at the nozzle exit p is set. Find the gas flow rate v from the nozzle as a function of the ratio p/p_c. Express your answer in terms of p_c, ρ_c, γ and the ratio p/p_c.	**2.00**

C2	At what value of p' does the outflow velocity from the nozzle reach its maximum value? What is this maximum speed $v_{\max}$? Express your answers in terms of p_c, ρ_c and γ.	**0.40**

C3	How does the nozzle outlet area A depend on the desired pressure at the nozzle outlet p (with fixed parameters at the nozzle inlet)? Those. find the function $A(p/p_c)$, assuming that the mass flow rate in each section of the nozzle is constant and equal to λ. Express your answer in terms of λ, p_c, ρ_c, γ and the ratio p/p_c.	**1.10**

C4	Find the pressure ratio p_t/p_c at the singular points for points B2 and B3. Express your answer in terms of the gas adiabatic index γ. Such pressures are called critical.	**1.50**

C5	For the nozzle shape defined in step B3, determine the dependence of the value v/c on the value p/p_c. Express your answer in terms of γ and p/p_c. Construct qualitative graphs of the dependence of the gas flow velocity v/c along the nozzle for the cases when the outlet pressure is greater than the critical one ($p/p_c > p_t/p_c$) and when the outlet pressure is less than the critical one ($p/p_c<p_t/p_c$).	**0.70**

C6	For the nozzle shape defined in point B3, the gas mass flow rate at the nozzle exit can be represented as: $$\lambda = C(A_{\text{вых}},\gamma,p_c,\rho_c) \cdot f(p/p_c),$$ где C — постоянная величина, зависящая только от A_{out}, γ, p_c и ρ_c, а $f(p/p_c)$ – функция, зависящая только от отношения p/p_c и $gamma$. Определите функцию $f(p/p_c)$. Ответ выразите через p/p_c и γ. Постройте качественный график массового расхода λ от давления на выходе из сопла ($p/p_c \in [0;1]$). Если на графике имеются особые точки, то укажите их на графике и получите соответствующие им аналитические выражения p/p_c.	**0.80**

Part D. Momentum acquired by a rocket (0.5 points)

Let the rocket now fly in the atmosphere. Atmospheric pressure p_0, gas pressure at the nozzle outlet p, nozzle area A, gas mass flow λ, gas flow rate from the nozzle v.

D1	Find the specific impulse acquired by the rocket while flying in the atmosphere. Express your answer in terms of p_0, p, A and v. Specific impulse is the additional impulse acquired by a rocket when releasing a unit mass of fuel.	**0.50**

Source: https://pho.rs/p/3040